Assessing the Value of Health IT Investment

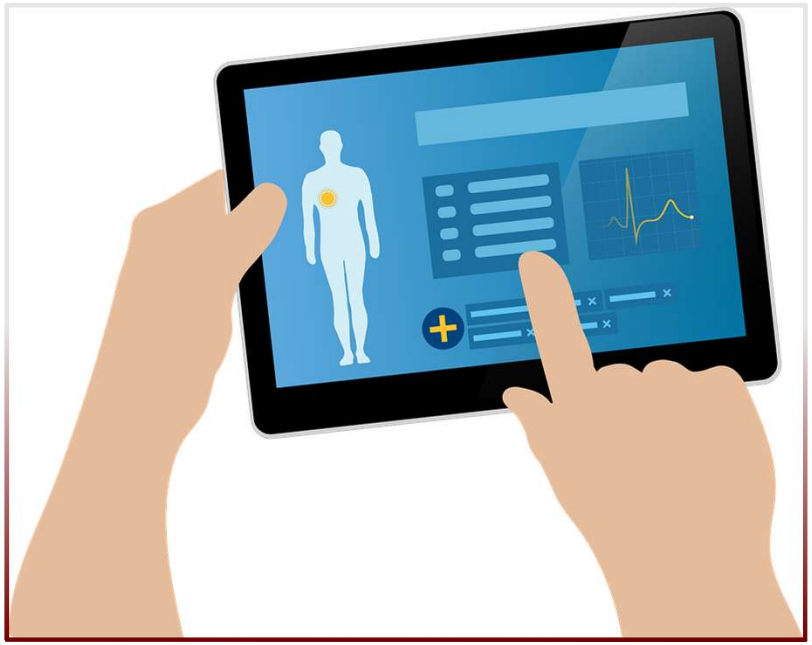
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HIE Healthcare Systems Governance Proposal

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Introduction

- **Context:** Data warehouse improvements created a stronger foundation for enterprise HIE.
- **Objective:** Test whether Epic improves communication across inpatient and ambulatory care.
- **Collaboration Value:**
  - **Connected record:** Gives providers across the 4-hospital network a clearer patient history.
  - **Seamless exchange:** Replaces disconnected inpatient and outpatient interfaces that delayed care.
  - **Fewer handoffs:** Cuts time spent locating, transferring, and reconciling data.
  - **Expected Benefits: Fewer discrepancies, duplicate tests, missed follow-up, readmissions, and communication costs (Menachemi et al., 2018).**
- **Implementation Challenge:** Adoption depends on reliable interoperability, easy access, and usable workflows.



HIE Capabilities

A single exchange layer turns a fragmented workflow into connected care.

TODAY

Limited integration

Ambulatory and hospital applications exchange information inconsistently, delaying access to clinical notes and patient data.

→

WITH EPIC

Standards-based exchange

HL7 and FHIR interfaces reduce custom point-to-point connections and support a common exchange layer.

→

CLINICAL VALUE

More timely information

Providers across the four-hospital network gain a clearer view of inpatient and ambulatory care.

Infrastructure models the HIE can be built on

CENTRALIZED

Central mainframe or cloud

All information is housed centrally and reached through departmental terminals, though exam-room lookups can lag.

DECENTRALIZED / FEDERATED

Data held at each node

Data stays at individual hospital or clinic nodes and is accessed via web or mobile apps, raising added security concerns.

HYBRID

Backbone plus local endpoints

A central enterprise backbone carries Master Patient Indexes while localized endpoints are queried via FHIR APIs and HL7 messaging.

Leadership implication: Epic provides the foundation; governance, identity matching, and interface monitoring sustain reliable exchange (Wager et al., 2021).

INFORMATION STANDARDS & DATA GOVERNANCE

Use the right exchange pattern—and manage security, traceability, and performance.

Information Standards

01 MESSAGING

HL7 v2 events

Supports event-driven clinical and administrative messages across implemented interfaces.

Best fit: ADT, orders, and results

02 RESOURCE ACCESS

FHIR R4 APIs

Standardizes resource interactions over HTTP. Each server declares the resources and operations it supports.

Not a guarantee of instant exchange

03 ASSURANCE

Governance controls

Apply authentication, authorization, TLS, provenance, audit logging, and operational monitoring.

Set latency, availability, and error targets

LEADERSHIP DECISION

Approve the interface profile, security controls, audit requirements, and service-level targets.

Sources: HL7 Version 2 Product Suite; HL7 FHIR R4 (Health Level Seven International, 2019); TEFCA v2.1 (ONC & The Sequoia Project, 2024).

Interoperability Challenges

LEADERSHIP TAKEAWAY

Epic provides the common platform; disciplined controls make exchange dependable (ONC & The Sequoia Project, 2024).

01 IDENTITY

Match the right patient

Reconcile legacy records and govern exceptions in the master patient index.

Proof: duplicate and unresolved-match trends

02 DATA

Preserve clinical meaning

Map source fields and terminology; validate migrated and exchanged content.

Proof: reconciliation results and defect trends

03 PERFORMANCE

Sustain reliable exchange

Monitor interfaces, latency, and errors; define ownership and escalation paths.

Proof: threshold compliance and recovery time

LEADERSHIP DECISION

Assign accountable owners and acceptance thresholds before go-live.

VALUE REALIZATION PLAN

Three gates convert technical readiness into measurable clinical value.

Participation Barriers

01 BARRIER

Adoption friction

Clinicians may bypass or underuse Epic when workflows feel unfamiliar or unreliable.

02 INTERVENTION

Build confidence

Use role-based scenarios drawn from hospital and ambulatory workflows; capture issues and retrain.

03 PROOF

Validate before launch

Simulate an ED transfer and confirm identity, active orders, notes, allergies, medications, and update timing.

MEASUREMENT DISCIPLINE

Adoption
Baseline → target → accountable owner

Reliability
Data completeness + latency threshold

Outcomes
Metric + measurement window + owner

Investment and payback figures require an approved-source baseline before inclusion (Dupont et al., 2023).

Recommendations (1 of 2)

Four owned actions create a repeatable listen → architect → measure → resolve operating model.

01 LISTEN

Optimize with clinicians

Collect structured feedback on workflow, usability, communication, and documentation burden.

Owner: Clinical Operations
Cadence: monthly

02 ARCHITECT

Adopt a hybrid HIE model

Balance centralized identity governance for accurate patient matching across the four hospitals with decentralized API access at ambulatory care points.

Benefit: local speed with enterprise security, audit logging, and interoperability

03 MEASURE

Track value realization

Report adoption, operational outcomes, and validated financial impact against the approved business case.

Owner: IT + Finance
Cadence: monthly; quarterly executive review

04 RESOLVE

Govern issues and vendors

Maintain one issue queue, prioritize clinical and revenue-cycle risk, and escalate against contracted service levels.

Owner: IT + Vendor Management
Cadence: weekly triage

LEADERSHIP DECISION

Approve the governance owners, initial scorecard, and review cadence.

Scorecard definition: baseline + target + owner + source + cadence + attribution boundary (Wager et al., 2021).

Recommendations (2 of 2)

Metrics that prove value for hospitals, patients, and providers

HOSPITALS

Post-implementation KPIs

Duplicate test rate, 30-day readmissions, interface error and latency thresholds, validated cost avoidance against the approved business case.

PATIENTS

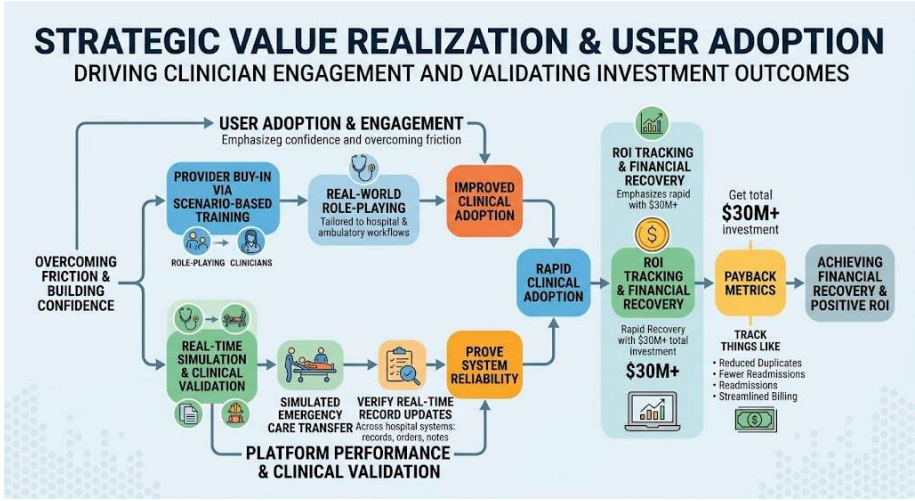
Testing feedback and validation

Identity match accuracy, record completeness at transfer, and update timing confirmed through simulated ED care-transfer testing.

PROVIDERS

Provider use surveys

Active-use rate, usability and satisfaction scores, chart-search time, and self-reported documentation burden.



Each measure carries a baseline, target, owner, source, and cadence before value is claimed (Dupont et al., 2023).

Conclusion

LEADERSHIP CONCLUSION

One platform. Trusted exchange. Consistent use.

01 CONNECT

Unify the care record

A shared inpatient and ambulatory platform gives providers a more complete view across the network.

Outcome: more coordinated decisions

02 TRUST

Control exchange risk

Identity governance, validated mapping, security, and interface monitoring make data dependable.

Outcome: safer, more reliable exchange

03 SUSTAIN

Turn use into value

Role-based adoption, measurable targets, and closed-loop governance keep improvement active after launch.

Outcome: accountable value realization

LEADERSHIP COMMITMENT

Keep owners accountable, review the scorecard, and remove barriers quickly (Menachemi et al., 2018; Dupont et al., 2023).

References

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